

Unary Iterator

$isOdd : P \rightarrow \{F, T\}$

$W = \{\text{false}, \text{true}\}$

$c = \text{true}$

$T = \lambda x. \neg x$

No fixed left input

$n = 1$

T

$n = 2$

F

$n = 3$

T

$n = 4$

F

$n = 5$

T

$f(1) = \text{true}$

$f(S(n)) = \neg f(n)$

One invocation of PIT

Key: W need not be P

Target set is $\{F, T\}$.

Iterator theorem fires

for any W, c, T .

Zero fixed inputs.

Binary Iterator

$2 + n$ (fix $m = 2$)

$W = P$

$c = S(2) = 3$

$T = S$ (successor)

One fixed left input m

$n = 1$

3

$n = 2$

4

$n = 3$

5

$n = 4$

6

$n = 5$

7

$f_m(1) = S(m)$

$f_m(S(n)) = S(f_m(n))$

One invocation per fixed m

Key: m baked into c

Left input m fixes start.

Right input n counts

T -steps applied.

One fixed input.

Ternary Iterator

$2 \uparrow \uparrow n$ (tetration, fix $m=2$)

$W = P$

$c = m = 2$

$T = \lambda x. m^{\wedge}x = 2^{\wedge}x$

Two fixed inputs m, base

$n = 1$

2

$n = 2$

4

$n = 3$

16

$n = 4$

65536

$n = 5$

2^{65536}

$f(1) = m$

$f(S(n)) = m^{\wedge}f(n)$

$2 \uparrow \uparrow n = \text{tower of } n \text{ } 2\text{s}$

Key: same machine

$T = (x \mapsto m^{\wedge}x)$ uses
exponentiation already
proved. Dependency: exp.

Two fixed inputs.

All three are configurations of (W, c, T) . The arity of the output operation = number of fixed inputs.